

EvidenceGate

A fail-closed evidence integrity system for live multimodal rail-port decision support

WORKING SOFTWARE PROTOTYPE + REPRODUCIBLE EXPERIMENT

1,600	100%	0
controlled trials	expected-state agreement	false READY outcomes in adverse trials

Submission field	Complete before upload
Student 1	Name: _____ Grade: _____
Student 2 (optional)	Name: _____ Grade: _____
School	School / City / State: _____
Guide	Name / Role: _____
Experiment executed	2026-08-31 11:16:01.217503 UTC

Scientific integrity note

This report contains measured software-validation data generated from the included scripts. It does not claim railway field validation, safety certification, authenticated freight telemetry or port-authority testing. Students should rerun and understand every experiment before submission.

Abstract

EvidenceGate is a working decision-support prototype designed to prevent unsupported rail-port logistics decisions from appearing operationally ready. The system binds route decisions to source identity, observation time, freshness, required evidence roles, checksums, HMAC-signed lineage and human approval. A controlled experiment executed 1,600 trials across eight scenarios. All 200 complete baselines returned READY; all 1,400 adverse trials returned HOLD or HARD_BLOCKED as specified. The adverse-case false-READY count was 0. A separate short live-provider observation made ten public API requests: five to Open-Meteo for Mumbai weather and five to NOAA SWPC for planetary Kp data. All ten returned HTTP 200 with the required schema. These results demonstrate deterministic fail-closed behavior under the tested software conditions, but do not establish railway safety, long-term provider uptime or real freight-field performance.

Keywords

evidence integrity; fail-closed software; data provenance; freshness; logistics; railway-port coordination; reproducible experiment

Original contribution

- A four-state evidence decision model: READY, HOLD, UNAVAILABLE and HARD_BLOCKED.
- A signed SourceLine record binding decision inputs, timestamps, transformations and data-source identity.
- Atomic multi-leg journey dispatch after fresh evidence verification and human approval.
- Truthful provider degradation: missing or malformed live data is shown as unavailable rather than replaced by a convincing fixed value.

INSEF fit

The current INSEF 2026-27 page accepts practical engineering or research projects in Computer Science & Engineering and Technology, for Grades 5-12 with up to two participants. It asks for a PDF containing project details and a project-demonstration video URL. This document supplies the technical PDF component; student identity and the final video URL must still be added during submission.

1. Problem statement and research question

Logistics dashboards often combine weather, port schedules, congestion estimates, engineering limits and operator declarations. If one source is stale, missing or silently replaced with a fixed fallback, a polished interface can communicate more certainty than the evidence supports. The engineering problem is therefore not only route calculation; it is preventing a result from being treated as dispatch-ready when its evidence cannot be verified.

Research question

Can a software evidence gate accept a complete, current and integrity-valid decision while consistently refusing decisions whose evidence is missing, stale, future-dated, seeded, hard-blocked or altered after signing?

Hypotheses

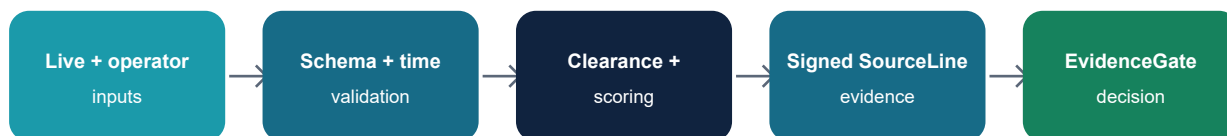
- H1: Complete baseline evidence will produce READY in every controlled trial.
- H2: Every tested adverse condition will prevent READY.
- H3: Post-signing modifications to snapshot or record trust fields will invalidate the signed root and produce HOLD.
- H4: Decision-gate evaluation will remain below 10 ms p95 on the test computer for these in-memory records.

Engineering objectives

- Remove operationally deceptive hardcoded live values and expose unavailable states.
- Validate provider schemas, observation timestamps, ranges and authorization contracts.
- Require certified engineering imports before static corridor limits can support READY.
- Bind schedules and approval receipts to the exact sealed evidence used for evaluation.
- Produce an incident-ready evidence kit with reproducible checksums and signer metadata.

2. Prototype architecture

EvidenceGate decision-support pipeline



READY only when required evidence is current and verifiable; otherwise HOLD, UNAVAILABLE or HARD_BLOCKED

Decision-state rules

State	Meaning	Operational interpretation
READY	Required evidence is available, current and integrity-valid.	Eligible for authorized approval and dispatch.
HOLD	Evidence exists but is stale, incomplete, untrusted or changed.	Re-evaluate or repair evidence; do not dispatch.
UNAVAILABLE	Required evidence or a signed snapshot is absent.	No decision can be made.
HARD_BLOCKED	Physical clearance or another absolute rule failed.	Numeric scores cannot override the block.

Live-data design

Open-Meteo supplies public weather when no OpenWeather key is configured. NOAA SWPC supplies planetary Kp observations. Authorized railway and maritime adapters require configured endpoints and bearer credentials. Their payloads must include bounded numeric values and timezone-aware observation timestamps. Configuration or provider failure produces UNAVAILABLE; it does not fall back to a passenger feed or fixed port window for operational scoring.

Evidence integrity

Each decision stores normalized records with source type, availability, observed and fetched times, freshness, value checksum and a checksum of the complete trust envelope. The snapshot and lineage graph are hashed into an evidence root and signed using HMAC-SHA256 with a key identifier. Verification uses the key identifier stored with the snapshot, permitting controlled key rotation while unknown historical signers fail closed.

Scope boundary

EvidenceGate is a research and decision-support prototype. It does not control trains, signalling, interlocking, braking, Kavach/ATP, port equipment or customs systems. Operational use would require railway and port authority integration, certified engineering datasets, cybersecurity review and regulatory approval.

3. Experimental method

Independent variable

The evidence condition was varied across eight controlled scenarios. Each scenario was executed 200 times with deterministic unique identifiers. The decision snapshot contained five required roles: clearance, weather, port alignment, congestion and historical delay.

Dependent variables

- Decision state returned by EvidenceGate.
- Agreement between expected and actual state.
- Evaluation latency measured with a monotonic high-resolution timer.
- Reason codes identifying why a decision was held or blocked.

Controls

- Same EvidenceGate 6.0.0 implementation and evidence-role set in all trials.
- Same cargo, route summary, weights and expected reliability score.
- Fixed experiment seed 20260831 and deterministic UUID generation.
- A process-local experiment-only HMAC key; no production secret was used or written to the results.
- Single computer and Python process to reduce environmental variation.

Scenario matrix

ID	Condition introduced	Expected
S1	Complete, fresh, signed baseline	READY
S2	Snapshot timestamp changed after sealing	HOLD
S3	Record source type changed after sealing	HOLD
S4	Weather decision role removed	HOLD
S5	Weather observation two hours old	HOLD
S6	Weather observation one hour in the future	HOLD
S7	Clearance evidence labelled seeded baseline	HOLD
S8	Physical clearance state hard-blocked	HARD_BLOCKED

Live-provider observation

Five sequential requests were made to each public provider from the same machine. Open-Meteo was queried at Mumbai coordinates 19.0760, 72.8777. A response counted as valid only if every scoring field was present. The NOAA response counted as valid only if the latest object row contained a timestamp and a numeric Kp value from 0 to 9. This is a short functionality check, not an uptime study.

4. Results

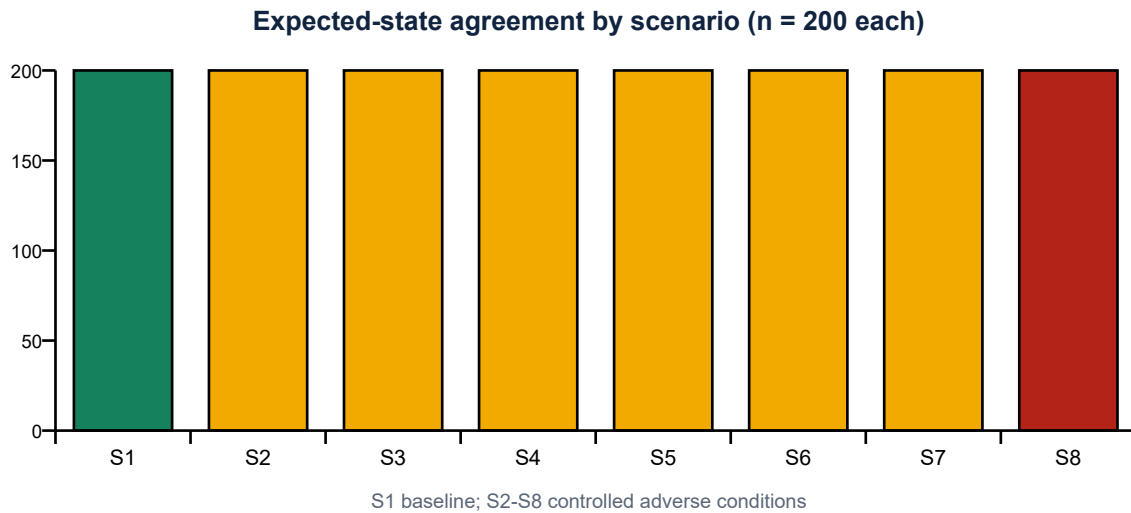


Table 1. EvidenceGate controlled experiment

Scenario	Actual state	Pass	Rate	Mean ms	p95 ms
S1 baseline ready	READY: 200	200	100.00%	1.725	1.916
S2 snapshot timestamp tamper	HOLD: 200	200	100.00%	1.681	1.821
S3 record source tamper	HOLD: 200	200	100.00%	1.795	2.048
S4 missing weather role	HOLD: 200	200	100.00%	1.434	1.659
S5 stale weather	HOLD: 200	200	100.00%	1.766	2.165
S6 future weather	HOLD: 200	200	100.00%	1.696	1.936
S7 seeded clearance	HOLD: 200	200	100.00%	1.686	1.873
S8 hard blocked	HARD_BLOCKED: 200	200	100.00%	1.663	1.814

All 1,600 trials matched their specified expected state. Baseline acceptance was 200/200. Across the 1,400 adverse trials, false READY outcomes were 0. The largest measured p95 decision-gate latency was 2.165 ms, supporting H4 on this machine and dataset.

Interpretation

H1-H4 were supported under the controlled conditions. Post-signing trust changes were detected by record-envelope and evidence-root verification. Missing and temporally invalid evidence produced HOLD. Seeded clearance could not qualify a route as READY. HARD_BLOCKED retained absolute precedence rather than being averaged away by a numerical reliability score.

5. Live-provider and system verification results

Table 2. Public-provider observations

Provider	Requests	HTTP 200	Schema valid	Availability*	Mean ms	p95 ms
Open-Meteo	5	5	5	100.0%	245.6	634.8
NOAA SWPC	5	5	5	100.0%	225.0	499.9

*Availability is limited to this five-request observation window and must not be interpreted as a long-term service-level estimate.

Observed result

All five Open-Meteo requests returned HTTP 200 and the required weather fields. All five NOAA SWPC requests returned HTTP 200 with a valid latest timestamp and Kp value. During fact-checking, NOAA's current object-row timestamp format was reproduced and the parser was updated to canonical UTC; a regression test now covers that format.

System release gate

Verification	Measured result
Backend automated suite	215 tests passed; repeated full gate
Backend static checks	Ruff and compileall passed; Alembic head 20260831_14
API contract	OpenAPI 6.0.0 exported; 58 paths
Web client	Type-check and lint passed; 197 modules built
Web dependency audit	No known production vulnerabilities reported
Android client	Unit tests, debug APK assembly and lint passed; 53 Gradle tasks
Release integrity	APK, web and source archives independently matched SHA-256 manifest

Unconfigured integrations

Supabase, authenticated freight rail, maritime berth, database and production evidence-signing credentials were not present in the execution environment. Therefore they were contract-tested with controlled payloads but not claimed as live production connections. The released software reports their absence explicitly and blocks READY where required.

6. Discussion

The experiment supports the central engineering claim: the prototype distinguishes a technically computed score from an evidence-qualified decision. A route can have a numerical score and still remain HOLD when a required input is stale or untrusted. This reduces the risk that interface polish or a plausible fallback value will be mistaken for operational evidence.

Why the signed evidence root matters

Checksumming only the displayed result would not protect source labels, observation time, transformation metadata or the decision graph. EvidenceGate includes these trust fields in record envelopes and includes the snapshot, record checksums and lineage edges in the signed root. The S2 and S3 trials show that changing a timestamp or source classification after sealing prevents READY.

Why fail-closed behavior matters

In a demo-oriented system, missing provider data is often replaced with a fixed number so the interface remains attractive. That behavior hides uncertainty. EvidenceGate instead communicates uncertainty as an operational state. This does not solve provider availability, but it prevents the software from silently converting absence into confidence.

Novelty and practical application

- Combines temporal freshness, provenance integrity and human approval in one dispatch gate.
- Makes provenance inspectable as a downloadable incident evidence kit rather than an internal log only.
- Applies atomic dispatch to multi-leg journeys so a waypoint chain is not partially dispatched through the main clients.
- Separates public environmental feeds, authorized enterprise feeds, certified imports, operator declarations and seeded baselines in the decision record.

Threats to validity

- The 200 repetitions per scenario exercise deterministic code with different identifiers; they are useful for consistency and latency observations but are not independent field events.
- The adverse scenarios are selected tests, not an exhaustive proof against all implementation or cybersecurity failures.
- The live-provider window is too short to estimate uptime, seasonal behavior or rate-limit performance.
- The experiment validates evidence-gate logic, not the accuracy of railway clearance limits, freight congestion or berth predictions.

7. Safety, ethics and data integrity

No fabricated evidence

No railway or maritime observation was invented to complete the report. Missing credentials are recorded as a limitation. Seeded engineering baselines remain labelled and are prevented from qualifying critical clearance as READY. Operator berth windows require a document reference and SHA-256 digest so the declaration is bound into the signed evidence.

Human responsibility

READY is not an autonomous authorization. The backend requires an authenticated operator, approver or administrator role and stores an approval receipt bound to the current evidence root. Any evidence-root change invalidates the old approval. This supports accountability but does not replace organizational procedures or legal authority.

Cybersecurity boundary

The prototype uses bearer-authenticated provider contracts, Supabase session verification, dedicated evidence-signing keys and retained verification-only keys for rotation. Production deployment still requires secret management, penetration testing, least-privilege network policy, audit retention, backups and incident-response exercises.

Responsible presentation

During judging, the prototype should be described as an evidence-integrity and decision-support system. It should not be described as an official Indian Railways dispatch system, a certified safety product, or a replacement for port, customs or railway authorities.

Student declaration before submission

- Rerun both experiment scripts and retain the newly generated CSV and JSON files.
- Understand and be able to explain every scenario, metric, limitation and code path shown in the demonstration.
- Complete the cover identity lines with accurate participant, grade, school and guide information.
- Add the final demonstration video URL to the submission form; do not insert a non-working link in this report.
- Disclose any mentor or AI assistance according to school and fair requirements.

8. Conclusion and future work

EvidenceGate 6.0.0 achieved 100% expected-state agreement across 1,600 controlled trials, with no false READY result in 1,400 adverse trials. The original public-provider check successfully validated five Open-Meteo and five NOAA SWPC responses during the 31 August 2026 observation window. These results support the prototype's fail-closed software behavior under the tested conditions.

Later verification, 7 September 2026: a separate controlled rerun again passed all 1,600 cases. NOAA returned five valid responses, but all five Open-Meteo requests timed out from the test host. These later observations are preserved separately under submission/experiments/runs and do not replace the original study data. Live availability is not guaranteed.

The experiment does not establish real-world railway safety or live freight-port accuracy. The next phase is field validation with authorized partners and certified data, followed by longer-duration provider monitoring and human-factors testing with intended operators.

Planned next experiments

Experiment	Required input	Primary metric
Certified clearance replay	Authority-approved segment limits	False-accept and false-block rate
Freight feed endurance	Authorized railway operations API	Availability, freshness and latency over 30 days
Berth alignment validation	Port-authority schedules and actual events	Arrival-window error and missed-window rate
Operator usability	Consent-based participant study	Decision time, comprehension and unsafe-action prevention
Security assessment	Independent test environment	Tamper, authorization and key-rotation findings

Final claim

EvidenceGate is a reproducibly tested software prototype that blocks unsupported readiness under the tested conditions. A live demonstration requires deployed services, real client configuration and device testing. Field validation and production railway certification remain outside these experimental results.

Appendix A. Reproducibility and raw evidence

Included experiment files

File	Purpose	SHA-256
run_evidencegate_experiment.py	Controlled 1,600-trial experiment	24971d2b322c85fc9c0cd8f219b68ee0fa11e50347c6ca835c18f4047db3d88f
evidencegate_experimental_results.csv	Raw trial-level observations	d10c0c2eca2213b5a3721964d80ad5d7979c129a850abd7b67466dcd27c0aa79
evidencegate_experimental_summary.json	Aggregate results and environment	97dab268e98c173f991612fee0c423c2df323157aaa9077f5fd589c71086ce83
run_live_provider_observations.py	Public-provider observation script	51fd7d1dc32b271ef845a01dc4e7d501eae3da6f076e4c0faf8edffde750bc4b
live_provider_observations.csv	Raw provider observations	acf1fc94e63c3abb88deed6cbe4a65cc6466fe71f066916154af30dbd226487d
live_provider_summary.json	Provider aggregates and limitations	e3680be85edc76883e9ad73fad236ed7f16bee07760cbadc638c6d5530eb7b9c

Repeat procedure

- Open PowerShell in the repository's backend directory.
- Set PYTHONPATH to a single dot for the current backend directory.
- Run .venv\Scripts\python.exe ..\submission\experiments\run_evidencegate_experiment.py.
- Run .venv\Scripts\python.exe ..\submission\experiments\run_live_provider_observations.py while connected to the internet.
- Compare the new aggregate state counts; provider latencies and observation values may legitimately differ.
- Run the full backend, web and Android verification commands documented in the repository before packaging.

Release SHA-256 manifest

```
16235d664d4ad468f3ca357b6cce87671635517ab4f86045781ba6fcac99747c EvidenceGate-v6.0.0-debug.apk
0ff458f958c4730d404b5eb96910d5af640b389932ddb679ba22d4ab6810c587 EvidenceGate-v6.0.0-source.zip
60518789974f608172bfcca8c30a6e3700eba504c770a01fcd1203469a56d22a EvidenceGate-v6.0.0-web.zip
17281eafc8b7570cd1a5e22d74c098a9df1c4c494d8f6151b71dbd2a1e56b022
backend/artifacts/models/delay_predictor/1.1.0-benchmark/metadata.json
114566c0e3be140b9d38ff9bb95d514a0a46bec92de98811271e1faa872eb0d1
backend/artifacts/models/delay_predictor/1.1.0-benchmark/model.joblib
```

Appendix B. References and INSEF upload checklist

References

- Science Society of India. INSEF 2026-27 Project Submission. <https://insef.org/insef/> (accessed 31 August 2026).
- Open-Meteo. Weather Forecast API documentation. <https://open-meteo.com/en/docs> (accessed 31 August 2026).
- NOAA Space Weather Prediction Center. Planetary K-index product feed. <https://services.swpc.noaa.gov/products/noaa-planetary-k-index.json> (accessed 31 August 2026).
- Krawczyk, H., Bellare, M., and Canetti, R. HMAC: Keyed-Hashing for Message Authentication. RFC 2104, 1997. <https://www.rfc-editor.org/rfc/rfc2104>
- EvidenceGate 6.0.0 source, evidence policy, deployment guide and experiment package included with this submission build.

Current INSEF upload checklist

Item	Status before upload
Eligible participant grade and category	Confirm Grade 5-12 and Junior/Senior rules
Maximum team size	Confirm one or two participants
Student, school and guide identities	Complete the cover-page identity lines
Project PDF	This visually verified report
Working demonstration	APK/web/source package available
Demonstration video URL	Record, upload and test access
Original logbook/raw data	Retain CSV, JSON, hashes and rerun notes
Regional submission	Submit to one region only by the current official deadline
Safety and assistance disclosures	Complete accurately with school/organizer guidance

Submission note: the official page currently lists 30 September 2026 for regional submissions and 25 October 2026 for Online-India, but participants must recheck the official page immediately before submission because dates and regional arrangements may change.